

# FE-TAD: Foundation Epistemic Tabular Anomaly Detection

Zero-Shot, Training-Free Anomaly Detection via In-Context Bayesian Predictive Uncertainty

## STEP 1: In-Context Support

(Zero Gradient Updates)

### Normal Reference Context

$\mathcal{D}_{\text{context}} \in \mathbb{R}^{n \times D}$  ( $n \approx 100$  normal rows)

- Historical unlabelled inliers
- Zero pseudo-anomalies needed
- Acts as conditioning prompt

### Unseen Test Instance

Query Vector  $x_i \in \mathbb{R}^D$

Case A: Normal Inlier (Conforms to data)

Case B: Outlier Anomaly (Violates priors)

## STEP 2: Tabular Foundation Model

(Self-Supervised Feature Projections)

### Self-Supervised Feature Projection

Target: Column  $c \in \mathcal{C}$  | Context: Columns  $\neg c$

Evaluates round-robin feature compatibility

### 12-Layer In-Context Transformer

Multi-Head Cross-Instance Attention

Implicit Kernel:  $k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$

### Continuous 999-Quantile Head

$\mathbf{q}(x_i) = [\hat{q}_{0.001}, \dots, \hat{q}_{0.999}]^\top$

Predictive Mean:  $\mu_c(x_i) = \frac{1}{999} \sum \hat{q}_\alpha$

Epistemic Variance:  $\sigma_c^2(x_i) = \frac{1}{999} \sum (\hat{q}_\alpha - \mu_c)^2$

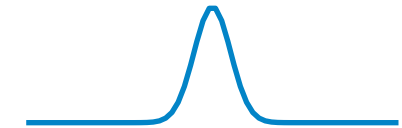
## STEP 3: Differential Entropy

(Epistemic Uncertainty Aggregation)

### Differential Epistemic Entropy

$$\mathcal{A}(x_i) = \sum_{c \in \mathcal{C}} \log(\sigma_c^2(x_i) + 10^{-4})$$
$$= \log \det \Sigma_{\text{epistemic}}(x_i)$$

Normal Inlier: Variance Collapses



Sharp posterior:  $\sigma^2 \rightarrow 0 \Rightarrow \mathcal{A}(x_i) \ll 0$

Anomaly: Variance Explodes



Dispersed posterior:  $\sigma^2 \gg 0 \Rightarrow \mathcal{A}(x_i) \gg 0$